

Spine Formal-v6 Primary Dynamic-Graph Results

Correctness-Gated Spine vs. GraSU+ReGraph K4-shared

Chuxiao Han

Live snapshot, July 2026

Abstract

This live report contains 16 of 34 expected formal-v6 executions and 3 complete Spine/K4-shared pairs. Missing bars are still-running or queued executions, never zero-valued measurements. Every admitted row passes architecture-precision and independent mathematical oracles, full final-state comparison, and request, response, and DRAM conservation.

1 Measurement contract

Algorithm	System	Window	Warm state
CC	G+R K4-shared	complete_execution	False
CC	Spine	complete_execution	False
ResPR	G+R K4-shared	complete_execution	False
ResPR	Spine	complete_execution	False
SSSP	G+R K4-shared	complete_execution	False
SSSP	Spine	dynamic_update_only	True

Table 1: Measurement windows observed in admitted formal-v6 rows.

SSSP interpretation. Spine starts from a verified persisted old-graph SSSP state and measures the accepted update, automatic active-source discovery, incremental propagation, and drain. The conversion-free GraSU+ReGraph baseline performs its declared complete ReGraph execution because it has no equivalent persisted incremental SSSP state. This is an architecture-level dynamic-service comparison, not an identical-kernel microbenchmark.

Claim boundary. The report is partial until all expected rows finish. Device cycles, accepted memory bytes, and DRAMSim3 HBM energy are simulator outputs. They do not claim cycle-for-cycle FPGA calibration, on-chip dynamic energy, or total board power.

2 Current primary comparison

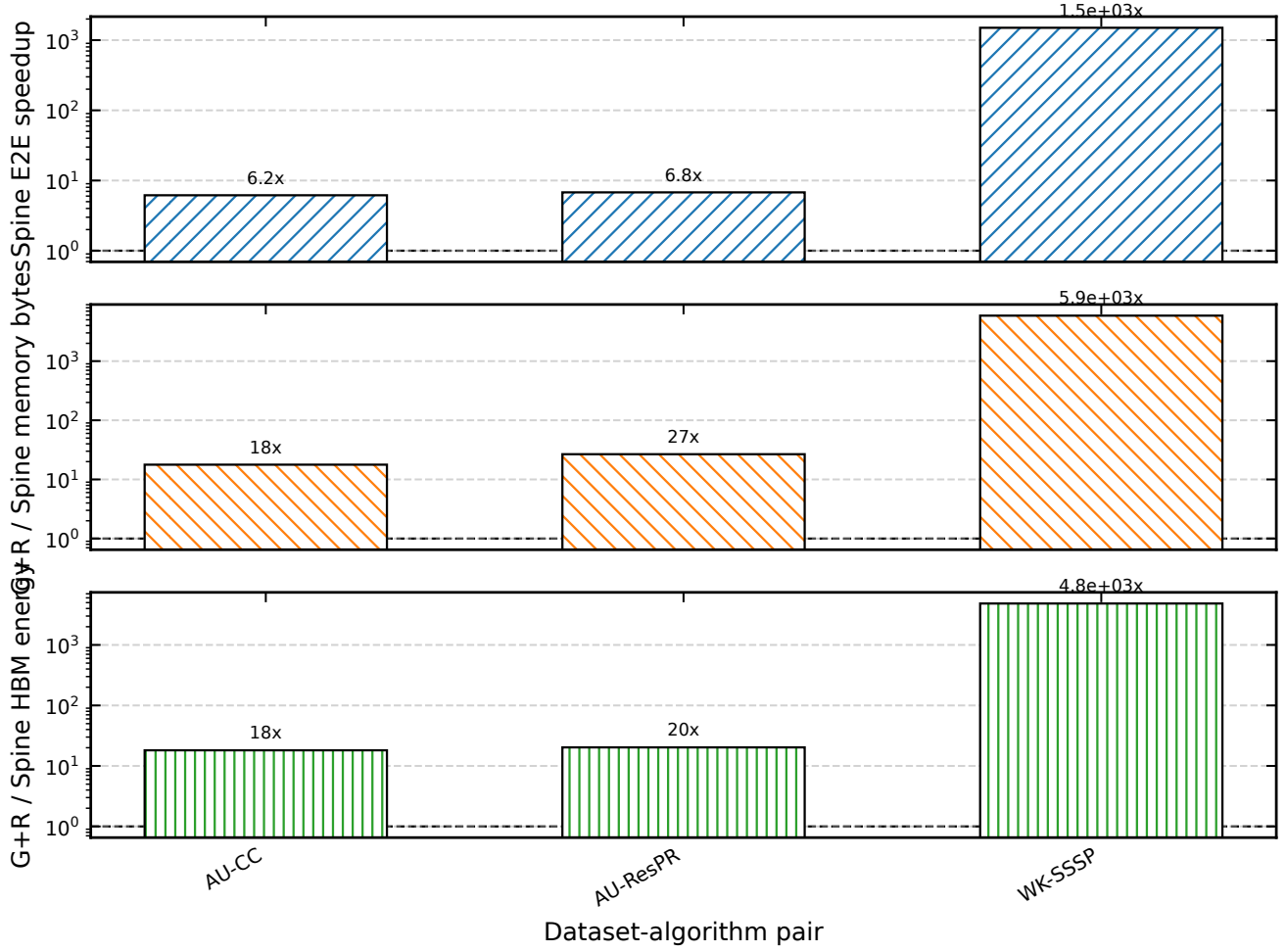


Figure 1: Ratios for complete insertion-batch-8 pairs. Values above one favor Spine for E2E latency and indicate that GraSU+ReGraph uses more memory traffic or HBM energy in the other panels.

Data	Algorithm	Spine cyc	G+R K4 cyc	Speedup
AU	CC	2,183,489	13,462,032	6.17×
AU	ResPR	2,172,364	14,729,898	6.78×
WK	SSSP	43,227	64,837,217	1499.92×

Table 2: Absolute device cycles. Only cross-system final-state-matched pairs enter this table.

3 Memory traffic and request locality

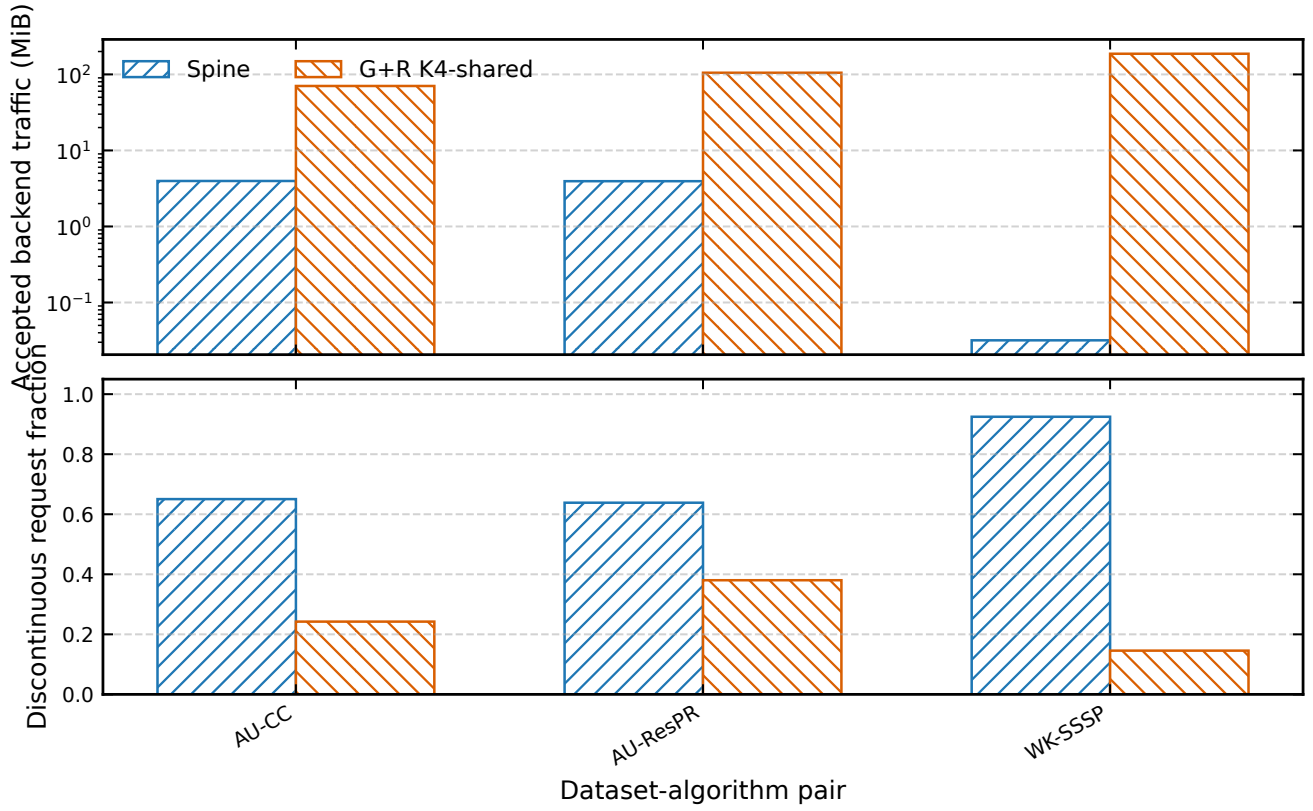


Figure 2: Absolute accepted-backend traffic and request-stream locality for the same complete pairs. A request is discontinuous when its accepted address does not continue the previous request from the same initiator, operation, and logical HBM channel. This is not a DRAM row-buffer-miss metric.

4 Update-only throughput

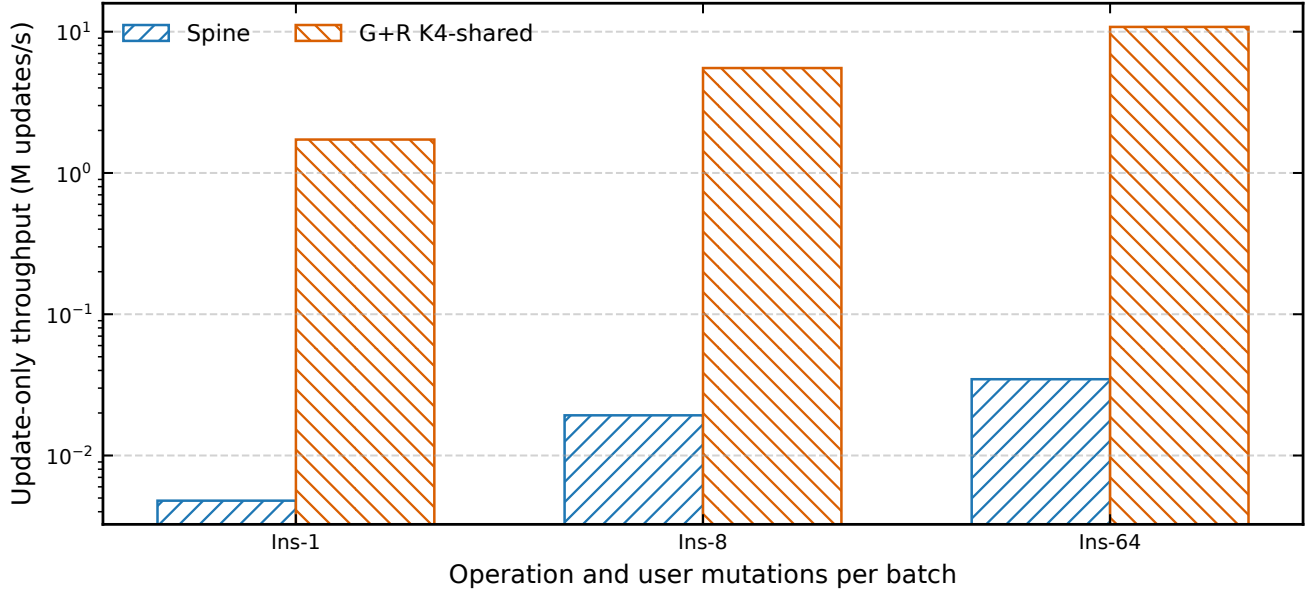


Figure 3: Correctness-gated AskUbuntu weighted-SSSP update-phase throughput. The current snapshot contains 3 of 9 planned operation–batch pairs. Ins uses the full materialized graph; Del and Wgt use the declared 64K-edge non-monotonic fallback slice. Missing points are not zeros.

Window boundary. This figure isolates graph-structure maintenance cycles and therefore exposes GraSU’s PMA update advantage. It must not be read as end-to-end dynamic graph service latency, which also includes differential discovery and propagation.

5 Implementation-level power attribution

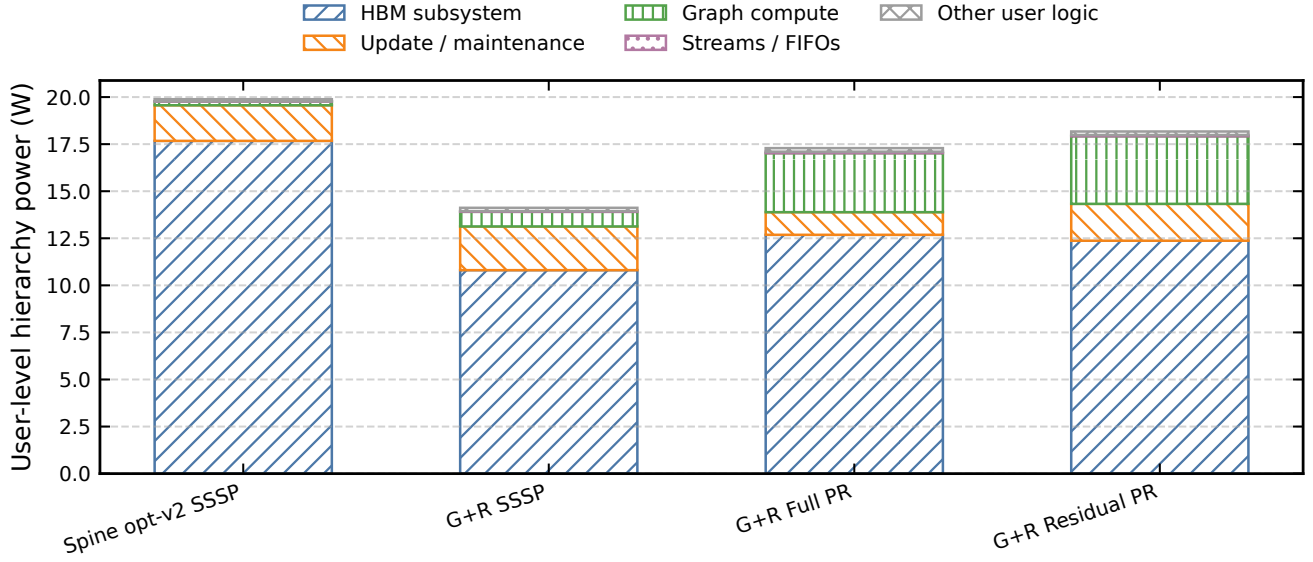


Figure 4: Vivado post-route vectorless hierarchy power at 150 MHz. The four bars are distinct routed builds; the Spine evidence covers SSSP only. Values use Vivado default activity with Low confidence and establish component attribution, not workload-calibrated energy or board power.

Energy boundary. The workload-specific HBM energy ratios in Figure 1 come from DRAMSim3 and are paired with the exact executions plotted there. The vectorless powers above must not be multiplied by the v6 latency to claim total workload energy. Workload-calibrated on-chip dynamic energy remains open because equivalent per-event energy models and complete activity counters are not yet available for every routed algorithm build.